LESSON 5.3

Determine the Area of Trapezoids

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a trapezoid using the formula A = ½(b1 + b2) × h.

WHY THIS MATTERS Your architecture firm is designing a trapezoidal window for a modern building. The glass supplier charges by the square foot, so the team must calculate the exact area. The window has a top edge of 4 feet, a bottom edge of 8 feet, and a height of 5 feet.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Trapezoid *Spanish: Trapecio*	A four-sided shape with just one pair of parallel sides.	top base and bottom base
Parallel *Spanish: Paralelas*	Two lines or sides that stay the same distance apart and never meet.	the two bases
Base 1 (b1) *Spanish: Base 1 (b1)*	One of the two parallel sides of a trapezoid.	If the bottom of the trapezoid is 10 ft, then b_1 = 10 ft in the formula A = ½(b_1 + b_2) × h
Base 2 (b2) *Spanish: Base 2 (b2)*	The other parallel side of a trapezoid.	If the top of the trapezoid is 6 ft, then b_2 = 6 ft; both bases are parallel to each other
Height *Spanish: Altura*	The straight-up distance between the two parallel sides.	h, meets bases squarely
Perpendicular *Spanish: Perpendicular*	Two lines that meet to make a square corner (90 degrees).	The dashed height meets the base at a square corner — the small 90° box is what makes it perpendicular, not slanted.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Bases 6 cm and 14 cm, height 5 cm, two matching triangles.

1. Rectangle: 6 × 5 = 30 cm².
2. Each triangle base: (14 − 6) ÷ 2 = 4 cm.
3. Triangles: 2 × (½ × 4 × 5) = 20 cm².

Answer: 30 + 20 = 50 cm²

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller trapezoid: bases 3 cm and 5 cm, height 2 cm.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The two parallel bases are _____ and _____ .
- I decomposed the trapezoid into _____ and _____ .
- The pieces have areas _____ , _____ , and _____ .
- The total area is _____ square units because _____ .

Word bank trapezoid | parallel | base | height | decompose | rectangle | triangle | leftover | add | total | area | square units

WATCH OUT!

A common mistake

A common mistake in Area of Trapezoids is forgetting the ½ in A = ½ × (b1 + b2) × h. After adding the bases and multiplying by the height, students stop there instead of taking half

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

What is the area of a trapezoid with bases 6 cm and 10 cm, and height 4 cm?

(A) 24 sq cm
(B) 32 sq cm
(C) 40 sq cm
(D) 64 sq cm

STEPS TO TRY

1. A trapezoid has two bases (6 cm and 10 cm) plus a height (4 cm) — all three belong in the formula.
2. Use A = ½ × (b_1 + b_2) × h: add the bases first, then multiply by the height, then take half.
3. Add 6 + 10 to get the base sum, multiply that by 4, then cut the result in half.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

2 **Circle the letter of the best answer. Show how you know.**

A trapezoid has an area of 45 sq ft, bases of 7 ft and 11 ft. What is the height?

(A) 3 ft
(B) 5 ft
(C) 9 ft
(D) 10 ft

STEPS TO TRY

1. You already know the area (45 sq ft) and both bases (7 ft and 11 ft) — the height is what's missing.
2. Simplify first: ½ × (7 + 11) = 9. So the equation becomes 45 = 9 × h.
3. Divide 45 by 9 to find the height in feet.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

What is the first step when finding the area of a trapezoid?

(A) Multiply the two bases
(B) Subtract the shorter base from the longer base
(C) Add the two bases together
(D) Divide the height by 2

STEPS TO TRY

1. Look at the trapezoid formula $A = ½ \times (b_1 + b_2) \times h$. Parentheses show what happens first.
2. Ask what $(b_1 + b_2)$ means: are the two parallel bases combined by adding, multiplying, or subtracting?
3. Say the formula in words: 'half of (base one ___ base two) times height.' The word that fills the blank is your first step.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A trapezoid has bases of 5 in and 9 in, and a height of 6 in. What is its area?

(A) 27 sq in
(B) 30 sq in
(C) 42 sq in
(D) 84 sq in

STEPS TO TRY

1. Circle the two bases (5 in and 9 in) and the height (6 in) — the trapezoid formula uses all three.
2. Follow $A = \frac{1}{2} \times (b_1 + b_2) \times h$ in order. Skipping the ½ makes your answer come out doubled.
3. Compute 5 + 9 = 14 first, then multiply 14 × 6, and finally take half of that product.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

EXPLAIN YOUR THINKING

If you copied this trapezoid, flipped it, and placed it next to the original, what shape would you get? How does this help you find the area?

SENTENCE STARTER The area of the trapezoid is 1/2 x (___ + ___) x ___ = 1/2 x ___ x ___ = ___ square feet.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help